

Emily Mader

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Berkeley Data Science master's student (Dec 2026) with experience building data and ML pipelines and open-source tools, applying Python, SQL, and statistics to scientific and manufacturing data.

SKILLS: Python (pandas, NumPy), SQL, R, Git, Azure, Microsoft Fabric (Lakehouse, Power BI) • ML: scikit-learn, XGBoost, TensorFlow • Statistics (OLS regression, time series, causal inference) • Visualization: matplotlib, seaborn, Bokeh, ggplot2

COURSEWORK: Applied Machine Learning • ML Systems Engineering (Docker, FastAPI) • Statistics for Data Science • Time Series Statistics • Fundamentals of Data Engineering • Experiments & Causal Inference

EXPERIENCE

Frone Systems – Data Analytics Intern 05/26 - 08/26

- Built and deployed an end-to-end Azure pipeline running unattended in production, ingesting manufacturing test data and automating cleaning, transformation, and summarization in Python to OneLake
- Delivered a Power BI dashboard on that pipeline surfacing failure modes and yield for the factory team
- Prototyped three ingestion architectures — event topic, storage queue, and change data feed — and drove design reviews, converging on the change data feed design
- Analyzed historical retest patterns in SQL and Python to quantify recovery rates by failure type and month
- Derived retest thresholds from a decision tree, revising team assumptions on which units to retest

A Round Group – Data Science Intern (part-time) 02/26 - 04/26

- Built a feature pipeline: loaded a MusicBrainz dump into local PostgreSQL and joined Census market data
- Developed XGBoost models for ticket demand and venue fill rate, tuned with Optuna; segmented artists with PCA/K-means and diagnosed feature-coverage limits constraining predictive validity

NASA Ames Research Center – OSTEM Intern, Machine Learning Research 06/25 - 08/25

- Replicated and evaluated a multi-output 1D CNN regression model on exoplanet spectral data, applying SHAP to extract feature importance and compare against physics-based methods
- Produced a feature-attribution methodology and recommended wavelength regions for telescope instrument design; presented findings to NASA instrument engineering teams

NASA Ames Research Center – OSTEM Intern, Scientific Software Development 06/24 - 12/24

- Designed modules, documentation, and tutorials for Xpectra, an open-source Python package for laboratory spectral data analysis; implemented data cleaning, regression, statistical analysis, and plotting utilities
- Processed large molecular line list files with parallelized multiprocessing; released beta on GitHub with docs

EDUCATION

Master of Information and Data Science | University of California, Berkeley Expected 12/26

- UC Berkeley School of Information Scholarship

B.S. in Physics (Astrophysics) | University of California, Santa Cruz 06/24

- Undergraduate research funding: Koret Scholarship (2024), Astronomy Award (2023)

SELECTED PROJECTS

Mars Terrain Classification: CNN terrain classification of Curiosity rover imagery with SAM2 misclassification

Energy Demand Forecasting: Regression and SARIMA models in R with evaluation and residual analysis

PUBLICATIONS

First Author: Uniform Analysis of Benchmark Brown Dwarfs | (Mader et al. 2026, The Astronomical Journal)

- Developed a Python workflow analyzing 130+ objects with MCMC uncertainty estimation, automated statistical tests, and visualizations; identified atmospheric trends and model systematics

Co-Author: Atmosphere of COCONUTS-2b | (Zhang et al. 2024, The Astronomical Journal)

PRESENTATIONS: NASA AI/ML Intern Showcase, 2025 (Talk) • STScI Habitable Worlds Observatory Symposium, 2025 • American Astronomical Society (AAS 243), 2024 • NASA Open Science Workshop, 2024